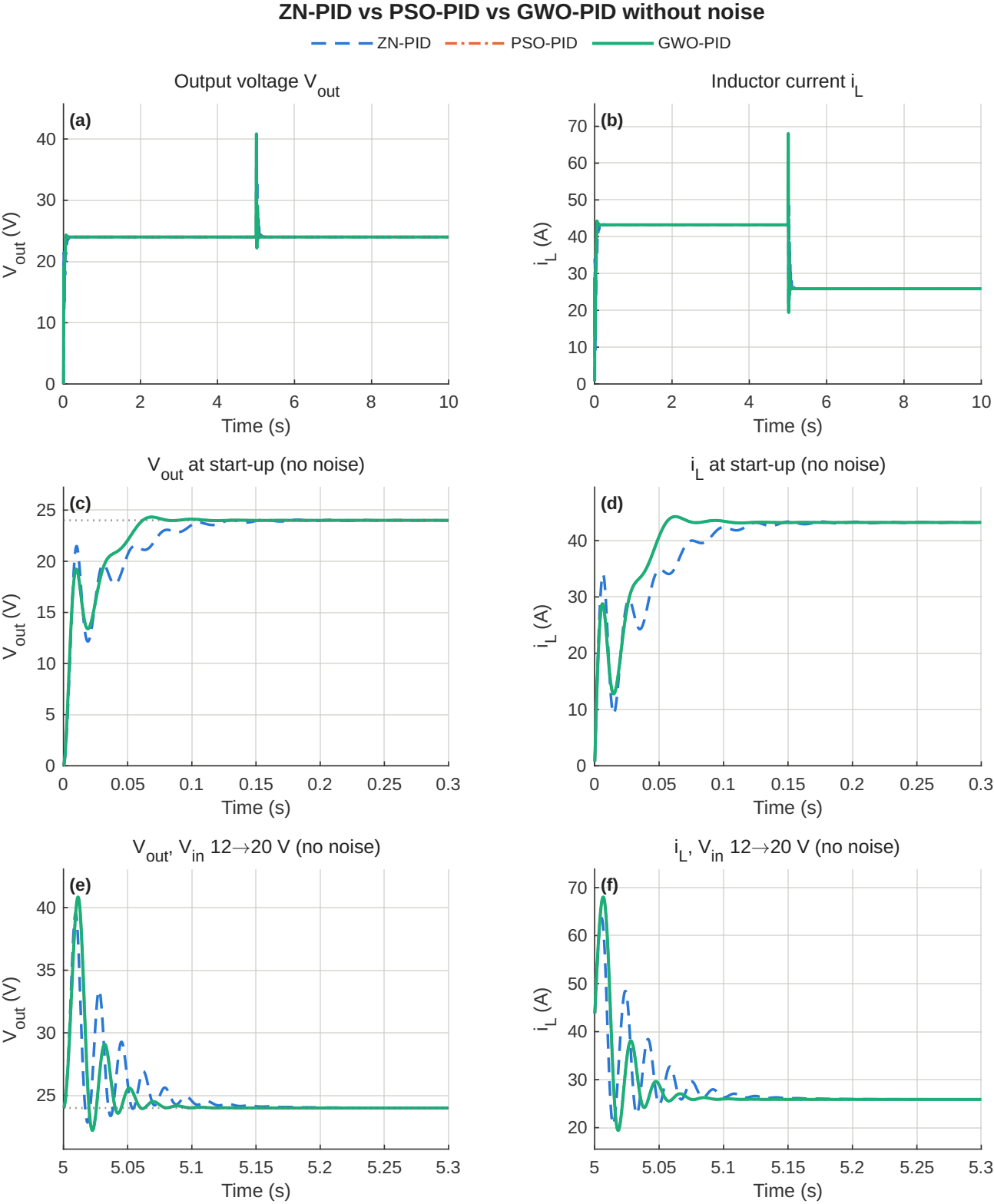

ZN-PID vs PSO-PID vs GWO-PID without noise
ZN-PID
PSO-PID
GWO-PID
Output voltage V_out
(a)
V_out (V)
Time (s)
Inductor current i_L
(b)
i_L (A)
Time (s)
V_out at start-up (no noise)
(c)
V_out (V)
Time (s)
i_L at start-up (no noise)
(d)
i_L (A)
Time (s)
V_out, V_in 12→20 V (no noise)
(e)
V_out (V)
Time (s)
i_L, V_in 12→20 V (no noise)
(f)
i_L (A)
Time (s)